

**Total Marks - 84****Attempt Questions 1 – 7****All questions are equal value.**

Answer each question in a SEPARATE writing booklet.

**Question 1** (12 Marks) Use a SEPARATE writing booklet.**Marks**

- (a) Find the exact value of  $\int_0^{\frac{\pi}{6}} \sec^2 2x dx$ . 2
- (b) Solve the inequation  $\frac{x}{x-3} \geq 2$ ,  $x \neq 3$ . 3
- (c) Find the acute angle between the lines  $x - y + 3 = 0$  and  $2x + y + 1 = 0$ .  
Give your answer correct to the nearest minute. 2
- (d) Find the co-ordinates of P, the point which divides the interval A(−5, −3) and B(4, −6) externally in the ratio 2 : 3. 2
- (e) When  $P(x) = x^3 + 3x^2 - mx + n$  is divided by  $(x + 2)$  the remainder is 9 and when  $P(x)$  is divided by  $(x - 3)$  the remainder is 49.  
Find the values of  $m$  and  $n$ . 3

**Question 2** (12 Marks) Use a SEPARATE writing booklet.

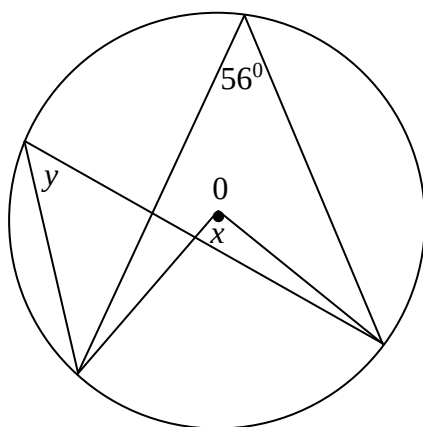
- (a) Sketch the graph of  $y = \cos^{-1}\left(\frac{x}{2}\right)$ . 2  
Your graph must indicate the domain and range.
- (b) Find  $\frac{d}{dx}(x \sin^{-1} 5x)$  2
- (c) (i) Write  $2 \cos \theta + \sin \theta$  in the form  $A \cos(\theta - \alpha)$  where  $A > 0$  and  $0 \leq \alpha \leq \frac{\pi}{2}$ . 2
- (ii) Hence, or otherwise solve the equation  $2 \cos \theta + \sin \theta = \sqrt{5}$  for  $0 \leq \theta \leq 2\pi$ . 2

Give your answer correct to 2 decimal places.

**Question 2 Continued.****Marks**

(d) Find the general solution for  $\cos \theta = \frac{\sqrt{3}}{2}$ . 2

- (e) O is the centre of the circle. Find the value of the pronumerals,  $x$  and  $y$ , giving reasons. 2

**Question 3** (12 Marks) Use a SEPARATE writing booklet.

- (a) Use the Binomial Theorem to find the term independent of  $x$  in the expansion of  $\left(x^2 - \frac{1}{2x}\right)^{18}$ . 3

(b) Given  $f(x) = \cos x - \ln x$

- (i) Show that a root of  $f(x) = 0$  lies between 0.5 and 1.5. 1

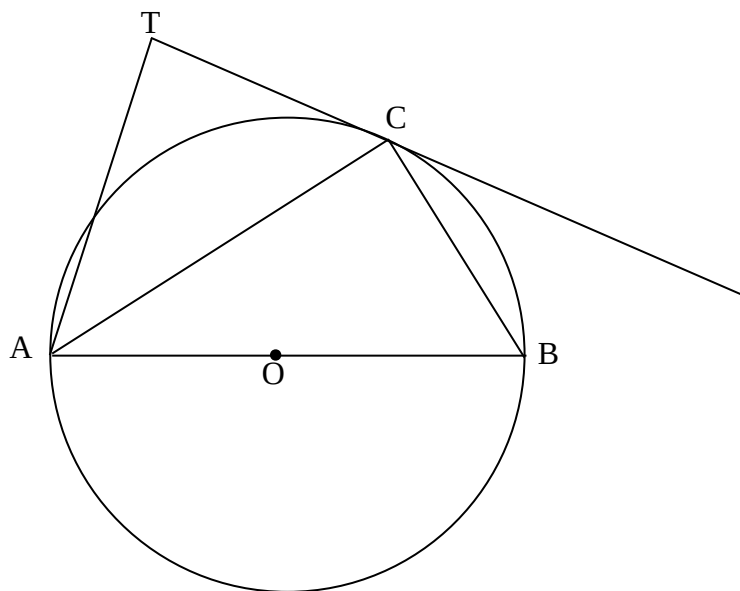
- (ii) Use Newton's Method once to find an approximation to this root of the equation, correct to 2 decimal places, starting with  $x = 1$ . 2

(c) Evaluate  $\int_0^{0.4} \frac{3dx}{4 + 25x^2}$  3

Questions continues on Page 3.

**Question 3 Continued.****Marks**

- (d) AOB is the diameter of a circle, centre O. C is the point of contact of the tangent, TC such that AC bisects  $\angle TAB$ . Prove that AT is perpendicular to TC. **3**



**Question 4** (12 Marks) Use a SEPARATE writing booklet.

- (a) Show by Mathematical Induction that  $9^n - 7^n$  is divisible by 2, for  $n \geq 1$ . **4**
- (b) How many different arrangements of the letters of the word NEWINGTON are possible? **2**
- (c) (i) A committee of eight people is to be formed from a group of twenty people. In how many different ways can the committee be formed? **1**
- (ii) This group consists of twelve men and eight women, how many ways can committee of four men and four women be formed if Kate and Pete must be included? **2**
- (iii) In how many ways can these 8 committee members sit around a circular table with no conditions restricting where anyone sits? **1**
- (iv) If the men and women alternate where they sit, how many arrangements are possible? **2**

**Question 5** (12 Marks) Use a SEPARATE writing booklet.**Marks**

- (a) P  $(2ap, ap^2)$  and Q  $(2aq, aq^2)$  are two points on the parabola  $x^2 = 4ay$ .

The tangents at P and Q meet at T which is always on the parabola  $x^2 = -4ay$ .

- (i) Given the equation of the tangent at P is  $y - px + ap^2 = 0$ . 2

Show that T is the point  $(a(p + q), apq)$ .

- (ii) Show that  $p^2 + q^2 = -6pq$ . 1

- (iii) Find M, the midpoint of PQ. 1

- (iv) Find the equation of the locus of M. 3

- (b) A casserole is cooling in a room that has a constant temperature of  $C$ . At a given time,  $t$  minutes the temperature decreases according to  $\frac{dT}{dt} = -k(T - 22)$  where  $k$  is a positive constant.

- (i) Show that  $T = 22 + Ae^{-kt}$  is a solution to the equation. 1

- (ii) Given that the initial temperature of the casserole is  $80^\circ\text{C}$  and it cools to  $60^\circ\text{C}$  after 10 minutes. Find A and  $k$ . 2

- (iii) How long will it take for the casserole to cool to  $30^\circ\text{C}$ ? 2

**Question 6** (12 Marks) Use a SEPARATE writing booklet.

- (a) Ben fires an arrow horizontally with a speed of  $60 \text{ ms}^{-1}$  from the top of a  $20 \text{ m}$  high cliff. Use  $g = 10 \text{ m/s}^2$ .
- (i) Show that the position of the arrow at time,  $t$  seconds is given by  $x = 60t$  and  $y = 20 - 5t^2$ . 2
- (ii) Find the time taken for the arrow to hit the ground. 1
- (iii) Find the distance the arrow is from the base of the cliff when it hits the ground. 1
- (iv) Find the acute angle to the horizontal at which the arrow hits the ground. 2
- (b) A particle's motion is defined by the equation,  $v^2 = 12 + 4x - x^2$  where  $x$  is its displacement from the origin in metres and  $v$  is its velocity in  $\text{ms}^{-1}$ . Initially, it is 6 metres to the right of the origin.
- (i) Show that the particle is moving in Simple Harmonic Motion. 1
- (ii) Find the centre, the period and the amplitude of the motion. 3
- (iii) The displacement of the particle at any time is given by the equation  $x = a \sin(nt + \theta) + b$ . 2
- Find the values of,  $\theta$  and  $b$  given  $0 \leq \theta \leq 2\pi$ .

**Question 7** (12 Marks) Use a SEPARATE writing booklet.

(a) Using the substitution  $u = 2x + 1$  find  $\int_0^1 \frac{4x}{2x+1} dx$  . **4**

(b) For  $(a + b)^n$  the general term is  $T_{k+1} = {}^nC_k a^{n-k} b^k$  .

For the following expansion  $(3 + 11x)^{19}$  :

(i) Show that  $\frac{T_{k+1}}{T_k} = \frac{11x(20-k)}{3k}$  . **2**

(ii) Hence find the greatest coefficient of  $(3 + 11x)^{19}$  . **3**

(c) (i) Use the binomial theorem to obtain an expansion for  $(1 + x)^{2n} + (1 - x)^{2n}$  **1**  
where  $n$  is a positive integer.

(ii) Hence evaluate  $1 + {}^{20}C_2 + {}^{20}C_4 + \dots + {}^{20}C_{20}$  . **2**

**END OF PAPER**

① a)  $\int_0^{\pi/6} \sec^2 2x dx = \left[ \frac{\tan 2x}{2} \right]_0^{\pi/6} = \frac{\tan \pi/3 - \tan 0}{2} = \frac{\sqrt{3}}{2}$  ① Correct Answer

b)  $\frac{x}{x-3} \geq 2 \quad \times (x-3)^2 (x-3) \geq 2(x-3)^2$

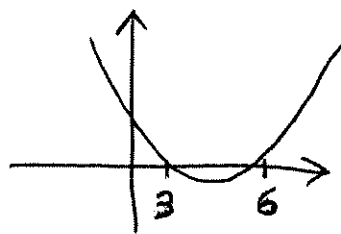
$x^2 - 3x \geq 2x^2 - 12x + 18$  ①

$0 \geq x^2 - 9x + 18$

$0 \geq (x-6)(x-3)$  ①

$3 < x \leq 6$

① Answer



c)  $\tan \alpha = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$x - y + 3 = 0$

$y = x + 3 \quad m_1 = 1$

$2x + y + 1 = 0$

$y = -2x - 1 \quad m_2 = -2$  ①

$\tan \alpha = \left| \frac{1+2}{1-2} \right| = 3$  ①

$\alpha = 71^\circ 33' 54'' \dots = 71^\circ 34' \text{ (nearest minute)}$

d) A (-5, -3) B (4, -6) External Division 2:-3

$xc = \frac{mx_2 + nx_1}{m+n} = \frac{8+15}{-1} = -23$  ①  $y = \frac{my_2 + ny_1}{m+n} = \frac{(2 \times -6) + (-3 \times -3)}{2-3} = \frac{-3}{-1} = 3$

P has co-ordinates (-23, 3). ①

e)  $P(x) = x^3 + 3x^2 - mx + n$

$P(-2) = 9$

$-8 + 12 + 2m + n = 9$

$2m + n = 5$  ① ①

$P(3) = 49$

$27 + 27 - 3m + n = 49$

$-3m + n = -5$  ② ①

①-②  $5m = 10 \quad \therefore m = 2 \quad \therefore n = 1$ . ①

### MARKERS COMMENTS

a) Generally well done. Some gave coefficient of 2 not  $\frac{1}{2}$

b) Most students knew to multiply by square of denominator. Mistakes came collecting terms, factoring & interpreting region. Students were not penalised for failure to recognise  $x \neq 3$ .

c) Many recalled the formula incorrectly

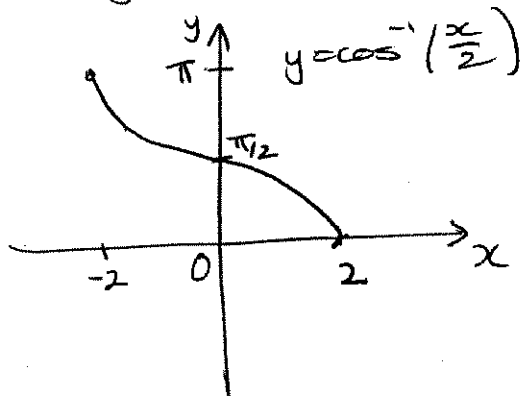
d) Students were poor at applying standard formula. Most common error was  $mx_1 + nx_2$  in numerator.

e) Students recognising factor theorem generally completed this question well.

(a) sketch  $y = \cos^{-1}\left(\frac{x}{2}\right)$

Domain  $-1 \leq \frac{x}{2} \leq 1$   
 $-2 \leq x \leq 2$

Range  $0 \leq y \leq \pi$



1 mark correct curve  
 1 mark correct domain  
 and range.

(b)  $\frac{d}{dx} (x \sin^{-1} 5x)$

$$= \sin^{-1} 5x + x \times \frac{5}{\sqrt{1-25x^2}}$$

$$= \sin^{-1} 5x + \frac{5x}{\sqrt{1-25x^2}}$$

1 mark correct differentiation  
 of  $\sin^{-1} 5x$

1 mark correct answer  
 from their working

(c)  $2\cos\theta + \sin\theta = A\cos(\theta - \alpha)$   $A > 0$   
 $0 \leq \alpha \leq \frac{\pi}{2}$

$$A = \sqrt{2^2 + 1^2}$$

$$A = \sqrt{5}$$

1 mark  $A = \sqrt{5}$

OR

$$\therefore 2\cos\theta + \sin\theta = \sqrt{5}\cos(\theta - \alpha)$$

and  $\tan\alpha = \frac{1}{2}$

$$\alpha = 0.463647609^\circ$$

$$\alpha = 0.46^\circ \text{ (2dp)}$$

1 mark  $\alpha = \tan^{-1}\left(\frac{1}{2}\right)$   
 or  $\alpha = 0.463647\dots$   
 $\alpha = \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$   
 $\alpha = \sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$

$\therefore 2\cos\theta + \sin\theta = \sqrt{5}\cos(\theta - 0.46^\circ)$  1 mark correct solution.

OR accept  $= \sqrt{5}\cos(\theta - \alpha)$   $\alpha = \tan^{-1}\left(\frac{1}{2}\right)$

degrees not accepted.

$$(c)(ii) 2\cos\theta + \sin\theta = \sqrt{5}$$

$$\therefore \sqrt{5}\cos(\theta - 0.46^\circ) = \sqrt{5}$$

$$\therefore \cos(\theta - 0.46^\circ) = 1$$

1 mark correct equation

$$\therefore \theta - 0.46^\circ = 0, 2\pi, 4\pi, \dots$$

$$\text{But } 0 \leq \theta \leq 2\pi$$

$$\theta = 0.46^\circ \text{ (2dp)}$$

1 mark correct  $\theta$

$$(d) \cos\theta = \frac{\sqrt{3}}{2}$$

$$\cos\theta = \cos\frac{\pi}{6}$$

general solution

$$\theta = 2n\pi \pm \alpha$$

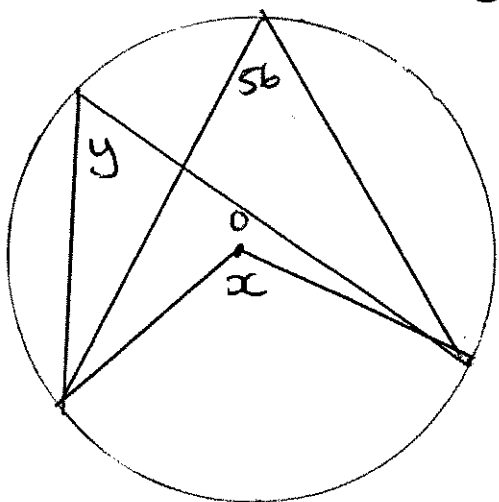
$$\theta = 2n\pi \pm \frac{\pi}{6}$$

$$\text{OR } \theta = -\frac{\pi}{6}, \frac{\pi}{6}, \frac{13\pi}{6}, \frac{23\pi}{6}, \dots$$

1 mark for  $\frac{\pi}{6}$

1 mark correct answer only

(e)



$$x = 2 \times 56^\circ \text{ (angle at the centre and the circumference)}$$

$$x = 112^\circ$$

$$y = 56^\circ \text{ (angles on the same arc)}$$

(angles on the same chord)  
(angles in the same segment)

1 mark each  
correct value with correct  
reason.

## Question 3

(a) Independent term  $\Rightarrow x^0$ 

$$\begin{aligned}
 (x^2 - \frac{1}{2}x^{-1})^{18} &\Rightarrow {}^{18}C_r (x^2)^{18-r} \left(-\frac{1}{2}\right)^r (x^{-r}) \\
 &= {}^{18}C_r x^{36-2r} \cdot x^{-r} \left(-\frac{1}{2}\right)^r \\
 &= {}^{18}C_r \left(-\frac{1}{2}\right)^r x^{36-3r}
 \end{aligned}$$

$$\therefore 36 - 3r = 0$$

$$\therefore r = 12$$

$$\therefore \text{Independent term} = {}^{18}C_{12} \left(\frac{1}{2}\right)^{12}$$

$$= \frac{4641}{1024}$$

- { 3 mks correct ans.  
 2 mks correct method for finding  $r$   
 1 mk correct expansion

(b) (i) For  $f(x) = \cos x - \ln x$  [1 mk] $f(x)$  is  $\downarrow$  for  $0.5 \leq x \leq 1.5$ 

$$f(0.5) > 0 \quad \text{ie.} \quad f(0.5) \approx 1.57$$

$$f(1.5) < 0 \quad \text{ie.} \quad f(1.5) \approx -0.33$$

 $\therefore$  root lies in the interval  $0.5 \leq x \leq 1.5$ 

(ii) By Newton's Method:

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \quad ; \text{ if } x_0 = 1$$

then  $x_1 = 1 - \frac{f(1)}{f'(1)}$

$$f'(x) = -\sin x - \frac{1}{x} \quad \therefore f'(1) = -\sin(1) - 1$$

$$f(x) = \cos x - \ln x \quad \therefore f(1) = \cos 1$$

$$\therefore x_1 = 1 + \frac{\cos 1}{\sin(1) + 1}$$

$$\therefore x_1 \approx 1.29 \quad (2 \text{ dp})$$

1 mk  
 { correct use of  
 formula  
 gives correct ans

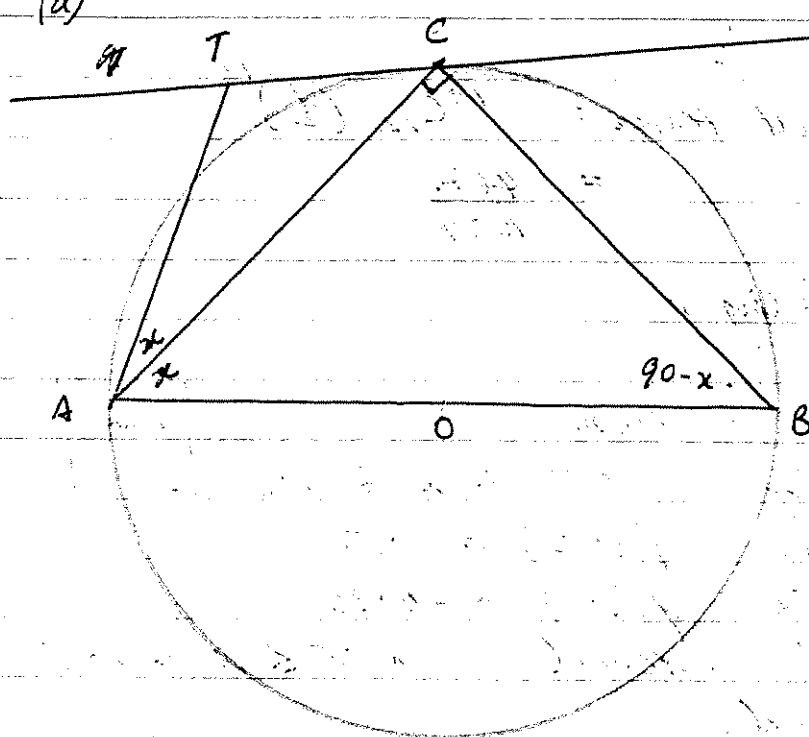
$$(c) \int_0^{0.4} \frac{3 \, dx}{4 + 25x^2} = 3 \int_0^{0.4} \frac{1}{4 + 25x^2} \, dx$$

$$= \frac{3}{25} \left[ \frac{25}{2} \cdot \tan^{-1} \frac{5x}{2} \right]_0^{0.4}$$

$$\left[ \begin{array}{l} 3 \text{ mks correct ans} \\ 2 \text{ mks correct integral} \\ 1 \text{ mks correct integration method} \end{array} \right] = 0.3 \left[ \tan^{-1}(1) - \tan^{-1}(0) \right]$$

$$= \frac{3\pi}{40}$$

(d)



TC is a tangent

AC bisects  $\angle TAB$ .

AOB is a diameter

$\angle ACB = 90^\circ$  (angle in a semi-circle).

$\therefore \angle ABC = 90 - x$ .

$\therefore \angle TCA = 90 - x$  (angle in alternate segment)

$\therefore \angle ATC = 90^\circ$  (angle sum of  $\triangle ATC$ )

$\left[ \begin{array}{l} 3 \text{ mks correct method} \\ 2 \text{ mks 1 mks (angle in semi)} \\ 1 \text{ mks (angle in alt seg)} \\ 1 \text{ mks (angles in } \triangle) \end{array} \right]$

#### Question 4

a) Step 1 ~~not~~ Prove true for  $n=1$

$$9^1 - 7^1 = 2 \therefore \text{divisible by } 2$$

Step 2 Assume true for  $n=k$

$$9^k - 7^k = 2M \text{ where } M \text{ is a positive integer}$$

Step 3 Prove true for  $n=k+1$

$$9^{k+1} - 7^{k+1} = 9 \cdot 9^k - 7 \cdot 7^k \quad \checkmark$$

$$= 7(9^k - 7^k) + 2 \cdot 9^k \quad \checkmark$$

$$= 7(2M) + 2 \cdot 9^k \text{ (From Step 2)}$$

$$= 2(7M + 9^k) \quad \checkmark$$

$\therefore 9^{k+1} - 7^{k+1}$  is divisible by 2

Step 4

b)  $\frac{9!}{3!} = 60480$

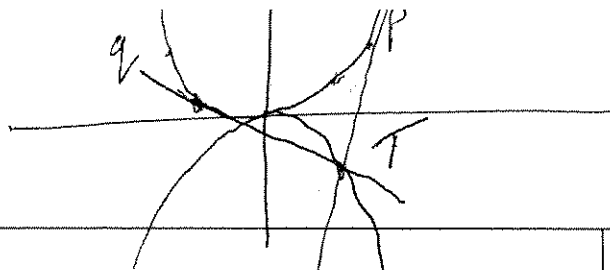
c) i)  ${}^{20}C_8 = 125970$

ii)  ${}^{11}C_3 \times {}^7C_3 = 5775$

iii)  $7! = 5040$

iv)

$$4! \times 3! = 144$$



| 5a) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marking Criteria                                                                                                                                                                                                                                                                                                                                         |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <p>(i) As the tangent at <math>p(2ap, ap^2)</math> is <math>y - px + ap^2 = 0</math> (1),<br/>therefore the tangent at Q is <math>y - qx + aq^2 = 0</math> (2).<br/>Solving (1) and (2) simultaneously:<br/>(2) - (1)<br/> <math display="block">(p - q)x - a(p^2 - q^2) = 0</math> <math display="block">\therefore x = \frac{a(p^2 - q^2)}{(p - q)}</math> <math display="block">= a(p + q)</math> <math display="block">\therefore y = p \times a(p + q) - ap^2</math> <math display="block">= apq</math> <math display="block">\therefore T(a(p + q), apq)</math> </p> <p>(ii) As T lies on <math>x^2 = -4ay</math><br/> <math display="block">\therefore (a(p + q))^2 = -4a \times apq</math> <math display="block">\therefore p^2 + 2pq + q^2 = -4pq</math> <math display="block">\therefore p^2 + q^2 = -6pq</math> </p> <p>(iii) Mid-point of PQ:<br/> <math display="block">M\left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2}\right) = M\left(a(p + q), \frac{a}{2}(p^2 + q^2)\right)</math> </p> <p>(iv)<br/>Let<br/> <math display="block">x = a(p + q), \quad y = \frac{a}{2}(p^2 + q^2)</math> <math display="block">\therefore y = \frac{a}{2}((p + q)^2 - 2pq)</math> <math display="block">\therefore y = \frac{a}{2}\left(\left(\frac{x}{a}\right)^2 - 2pq\right)</math> <math display="block">\therefore y = \frac{x^2}{2a} - apq</math> <math display="block">\therefore y = \frac{x^2}{2a} + \frac{a(p^2 + q^2)}{6}</math> <math display="block">\therefore y = \frac{x^2}{2a} + \frac{a(p^2 + q^2)}{2} \times \frac{1}{3}</math> <math display="block">\therefore y - \frac{y}{3} = \frac{x^2}{2a}</math> <math display="block">\therefore x^2 = \frac{4a}{3}y</math> <p>Locus of M is a parabola.</p> </p> | <p><b>Marking Criteria</b></p> <p><b>1 mark process</b></p> <p><b>1 mark for correct working</b></p> <p><b>1 mark for correctly using given information</b></p> <p><b>1 mark for correct working</b></p> <p><b>1 mark for starting with the correct parameters!</b></p> <p><b>1 mark for correct process</b></p> <p><b>1 mark for correct answer</b></p> |

| 5b) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marking Criteria                                                                                                                                                                                                  |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <p>Given <math>\frac{dT}{dt} = -k(T - 22)</math></p> <p>(i)</p> $T = 22 + Ae^{-kt}$ $\frac{dT}{dt} = -kAe^{-kt}$ $\frac{dT}{dt} = -k(T - 22) \quad \text{as} \quad Ae^{-kt} = T - 22$ <p>(ii) Now,<br/>when <math>t = 0</math>, <math>T = 80^\circ</math><br/> <math>\therefore 80 = 22 + A \times 1</math><br/> <math>\therefore A = 58</math><br/> And<br/> when <math>t = 10</math>, <math>T = 60^\circ</math><br/> <math>\therefore 60 = 22 + 58 \times e^{-k \times 10}</math><br/> <math>\therefore e^{-10k} = \frac{19}{29}</math><br/> <math>\therefore k = -\frac{1}{10} \ln \left( \frac{19}{29} \right)</math></p> <p>(iii)<br/>As<br/> <math>T = 22 + 58 e^{\frac{t}{10} \left( \ln \frac{19}{29} \right)}</math><br/> when <math>T = 30</math><br/> <math>\frac{4}{29} = e^{\frac{t}{10} \left( \ln \frac{19}{29} \right)}</math><br/> <math>\therefore \frac{t}{10} = \frac{\ln \left( \frac{4}{29} \right)}{\ln \frac{19}{29}}</math><br/> <math>\therefore t \approx 46.84 \text{ min} \quad \left( \begin{array}{l} 46 \text{ min} \\ 51 \text{ sec} \end{array} \right)</math></p> | <p><b>1 mark for correct working</b></p> <p><b>1 mark for correct value of A</b></p> <p><b>1 mark for correct value of k</b></p> <p><b>1 mark for correct process</b></p> <p><b>1 mark for correct answer</b></p> |

Q6

(i) Horizontally

$$\ddot{x} = 0$$

$$\dot{x} = t + c$$

$$\text{when } t=0, \dot{x}=60$$

$$\therefore \dot{x} = 60$$

$$\therefore x = 60t + c$$

$$\text{when } x=0, t=0$$

$$\therefore \boxed{x = 60t}$$

Vertically

$$\ddot{y} = -g = -10$$

$$\dot{y} = -10t + c$$

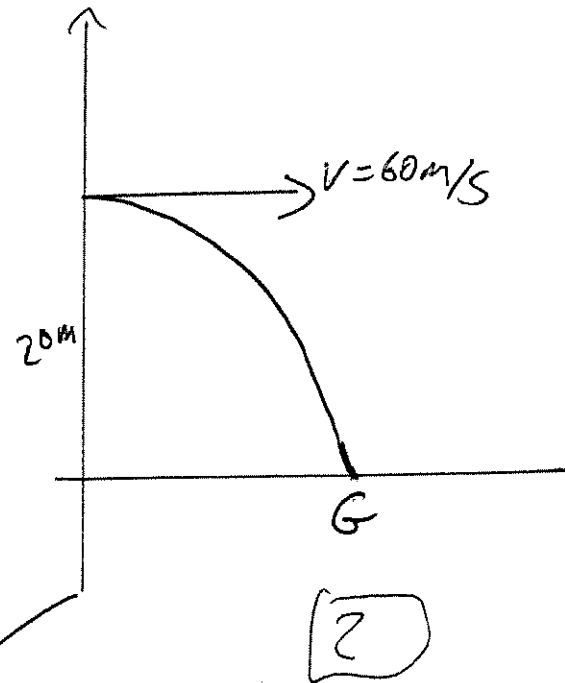
$$\text{when } t=0, \dot{y}=0$$

$$\therefore \dot{y} = -10t$$

$$\therefore y = -\frac{10t^2}{2} + c$$

$$\text{when } t=0, y=20$$

$$\therefore \boxed{y = 20 - 5t^2}$$

(ii) Arrow hits the ground when  $y=0$ 

$$\therefore 20 - 5t^2 = 0$$

$$(2-t)(2+t) = 0$$

$$\therefore \text{after 2 seconds } [t \geq 0] \quad \checkmark$$

1

(iii) when  $t=2$ 

$$x = 60 \times 2 = 120 \text{ m}$$

✓

1

(iv) at G

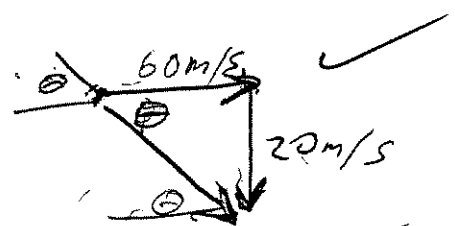
$$\dot{x} = 60 \text{ m/s}$$

$$\dot{y} = -20 \text{ m/s}$$

$$\tan \theta = \frac{20}{60}$$

$$\theta = \tan^{-1}\left(\frac{1}{3}\right)$$

$$\text{accept } \theta = \underline{1.8^\circ 26'} \quad \boxed{2}$$



(b)

$$v^2 = 12 + 4x - x^2$$

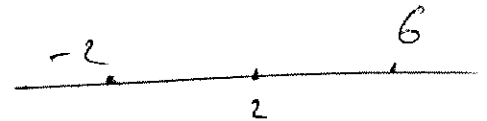
$$(i) \quad \frac{1}{2} v^2 = 6 + 2x - \frac{1}{2} x^2 \quad (1)$$

$$\frac{d(\frac{1}{2} v^2)}{dx} = 2 - x \quad \therefore \text{SHM as } \checkmark$$

$x$  is directly proportional to its displacement.

$$(ii) \quad v^2 = (6 - x)(2 + x)$$

$$v = \pm \sqrt{(6 - x)(2 + x)}$$



Centre of Motion:  $x = 2$

$$\text{Period} = 2\pi = 2\pi \text{ sec}$$

$$\text{Amplitude} = 4 \text{ m}$$

✓  
✓  
✓  
(3)

$$(iii) \quad x = a \sin(n\omega t + \theta) + b$$

$$x = 6, a = 4, n = 1, t = 0$$

$$6 = 4 \sin \theta + b$$

$$\text{now } \dot{x} = a n \cos(n\omega t + \theta)$$

$$\dot{x} = 0 \text{ when } t = 0$$

$$\therefore \cos \theta = 0$$

$$\theta = \frac{\pi}{2} \neq$$

(2)

$$\therefore 6 = 4 \sin \frac{\pi}{2} + b \quad \text{when } t = 0, x = 6$$

$$\therefore b = 2 \neq$$

Question:

$$(a) \int_0^1 \frac{4x}{2x+1} dx$$

$$\text{let } u = 2x+1$$

$$\therefore u-1 = 2x$$

$$\therefore \int_0^1 \frac{2x}{2x+1} \cdot 2 dx$$

$$1 du = 2 dx$$

$$\text{when } x=1 \quad u=3$$

$$x=0 \quad u=1$$

$$= \int_1^3 \frac{u-1}{u} \cdot du$$

$$= \int_1^3 \frac{u}{u} - \frac{1}{u} \cdot du$$

$$= \int_1^3 1 - \frac{1}{u} du$$

$$= u - \ln u \Big|_1^3$$

$$= (3 - \ln 3) - (1 - \ln 1)$$

$$= 3 - \ln 3 - 1$$

$$= 2 - \ln 3.$$

① mark

for correct manipulation.

① mark

correct substitution of their variables

① mark

correct integration of their expression as long as the expression is not easier.

① mark

correct answer only.

$$(b) (3+11x)^{19} = (a+b)^n$$

$$\text{then } T_{k+1} = {}^nC_k a^{n-k} \cdot b^k$$

$$(1) \text{ Show } \frac{T_{k+1}}{T_k} = \frac{11x(20-k)}{3k}$$

Using the rule

$$\frac{T_{k+1}}{T_k} = \frac{n-k+1}{k} \cdot \frac{b}{a}$$

$$= \frac{19-k+1}{k} \cdot \frac{(11x)}{3}$$

① mark  
correct substitution  
nb formula.

$$\therefore \frac{T_{k+1}}{T_k} = \frac{11x(20-k)}{3k}.$$

① mark  
correct manipulation

### Question 7

$$cc) (1+x)^{2n} + (1-x)^{2n} \quad n > 0$$

$$(1+x)^{2n} = \binom{2n}{0} + \binom{2n}{1}x + \binom{2n}{2}x^2 + \dots + x^{2n}$$

$$(1-x)^{2n} = \binom{2n}{0} - \binom{2n}{1}x + \binom{2n}{2}x^2 + \dots + (-x)^{2n}$$

$$\begin{aligned}(1+x)^{2n} + (1-x)^{2n} &= 2 \times \binom{2n}{0} + 2 \times \binom{2n}{2}x^2 + \dots + 2 \binom{2n}{2n}x^{2n} \\ &= 2 \left( \binom{2n}{0} + \binom{2n}{2}x^2 + \dots + \binom{2n}{2n}x^{2n} \right)\end{aligned}$$

① mark correct answer in any form.

### Evaluate

$$1 + {}^{20}C_2 + {}^{20}C_4 + \dots + {}^{20}C_{20}$$

$$= 1 \left( 1^{2n} + \binom{2n}{2}x^2 + \dots + \binom{2n}{2n}x^{2n} \right)$$

where  $n=10$   $x=1$   $\leftarrow$  ① mark correct substitution

$$= 1 \left( 1^{20} + {}^{20}C_2 1^2 + {}^{20}C_4 1^4 + \dots + {}^{20}C_{20} 1^{20} \right)$$

$$= 1 \left( 1 + {}^{20}C_2 + {}^{20}C_4 + \dots + {}^{20}C_{20} \right)$$

$$= \frac{(1+x)^{2n} + (1-x)^{2n}}{2} \quad \text{where } x=1 \quad n=10$$

$$= \frac{2^{20} + 0^{20}}{2}$$

$$= 2^{19}$$

① mark correct answer  
Must show where the answer is derived from using (i)

$$= 524288$$

(i) without the rule: Using  $T_{k+1} = {}^nC_k a^{n-k} \cdot b^k$   
 $T_k = {}^nC_{k-1} a^{n-(k-1)} \cdot b^{k-1}$   
 $T_k = {}^nC_{k-1} a^{n-k+1} \cdot b^{k-1}$

$$T_{k+1} = {}^{19}C_k 3^{19-k} \cdot (11x)^k$$

$$T_k = {}^{19}C_{k-1} \cdot 3^{19-k+1} \cdot (11x)^{k-1}$$

$$= {}^{19}C_{k-1} \cdot 3^{20-k} \cdot (11x)^{k-1}$$

① mark correct expression for  $T_k$

$$\therefore \frac{T_{k+1}}{T_k} = \frac{{}^{19}C_k \cdot 3^{19-k} \cdot (11x)^k}{{}^{19}C_{k-1} \cdot 3^{20-k} \cdot (11x)^{k-1}}$$

$$= \frac{19!}{k! (19-k)!} \cdot 3^{19-k} \cdot (11x)^k$$

$$\frac{19!}{(k-1)! (20-k)!} \cdot 3^{20-k} \cdot (11x)^{k-1}$$

$$= \frac{(k-1)! (20-k)! \cdot 11x}{k! (19-k)! \cdot 3}$$

$$= \frac{(20-k) \cdot 11x}{k \cdot 3}$$

$$= \frac{11x(20-k)}{3k}$$

Working must be shown.

(ii) Greatest coefficient occurs when  $\frac{T_{k+1}}{T_k} \geq 1$

$$\therefore \frac{11(20-k)}{3k} \geq 1$$

① mark for their result from (i) without the  $x$ .

$$11(20-k) \geq 3k$$

$$220 - 11k \geq 3k$$

$$220 \geq 14k$$

$$14k \leq 220$$

$$k \leq 15\frac{5}{7}$$

$$\therefore k = 15$$

$$\therefore T_{k+1} > T_k \text{ when } k \leq 15\frac{5}{7}$$

$$\therefore T_{16} > T_{15}$$

$$\therefore T_{16} \text{ has the greatest coefficient } T_{16} = {}^{19}C_{15} \cdot 3^4 \cdot 11^{15}$$

① mark correct answer.